

Part I: Measurement Cost of Generator Selection in Fixed-State ADAPT-VQE

[AUTHOR: names and affiliations]

September 2026

Abstract

This report defines and compares three measurement strategies for the generator-selection step in gradient-based ADAPT-VQE (adaptive derivative-assembled pseudo-Trotter variational quantum eigensolver): **M1: FC-UG**, fully commuting (FC) universal groups for static all-gradient estimation; **M2: BAI-FC-IG**, best-arm identification (BAI) using FC individual-gradient measurements; and **M3: BAI-FC-UG**, best-arm identification using fixed parent FC universal groups. A fourth construction, a *no-sharing baseline* that uses neither mechanism, completes the two-by-two the three methods span and makes each mechanism separately attributable.

Two classes of number are reported, and they are kept apart throughout. *Planning bounds* (Secs. 10–11) are oracle diagnostics: exact gradients and exact fragment variances are used to compute the shots each method would need, and the elimination rule reads exact values, so it cannot misfire. A *finite-shot surrogate* (Sec. 12) replaces the exact gradients with Gaussian estimates built from a finite number of shots and runs the same elimination rule on those. The surrogate samples estimators, not individual measurement outcomes, so it is not a shot-level simulation of a quantum device; the distinction and its consequences are set out in Sec. 7.

Results. On the planning bounds, M3 is cheaper than M1 in all nine cases, by factors of 1.1 to 192, and cheaper than M2 in eight of nine. The M2-versus-M3 comparison is *unresolved* on every H₂O row: the ratio moves over 0.74 to 1.30 across arbitrary choices of grouping routine, which spans unity. Against the no-sharing baseline, elimination alone is worth 1.3 to 253 and sharing alone only 1.1 to 3.4, and the two mechanisms are substitutes rather than complements: the interaction term is below unity in all nine cases. Under the finite-shot surrogate both best-arm methods cost more than their bounds, M2 by 2.2 to 3.2 and M3 by 1.4 to 3.0; an unseparated part of that is discretisation of the radius schedule rather than noise, and no mechanism for the difference between the two methods is established. Because M1 realises its bound exactly, the comparisons against M1 reorder under finite shots and M3 loses on two of seven rows. Every trial of the main finite-shot table returned the correct generator, which at 200 trials bounds the error rate only below 1.9%; the stated family-wise failure probability does not control that rate.

Scope. Every number is a single-ADAPT-step diagnostic for one fixed state, and counts shots only: circuit depth, the number of distinct measurement settings, and state-preparation cost are excluded (Sec. 7). Part III symmetry filtering and tapering are not included in the Part I tables; Part III supplies a smaller input problem to the same measurement methods. The revision history of this document, the naming convention for code and files, and the implementation status are in Apps. A–C.

1 Problem statement

Given a molecular Hamiltonian H , a current ADAPT-VQE state $|\psi\rangle$, and a unitary coupled-cluster singles-and-doubles (UCCSD) generator pool $\{G_i\}$, estimate the measurement cost required to identify a generator with the largest absolute gradient

$$i^* = \arg \max_i |g_i|, \quad g_i = \langle \psi | [H, G_i] | \psi \rangle. \quad (1)$$

The scientific question is whether using winner-identification statistics and shared fully commuting measurement contexts reduces the gradient-selection measurement cost relative to static all-gradient estimation.

The answer, in one sentence. Mostly yes: on oracle planning bounds at family-wise $\delta = 0.05$, combining both mechanisms (M3) costs 1.1 to 192 times less than static all-gradient estimation (M1) across the nine cases studied — but the change of statistical target carries the saving in seven of the nine, shared contexts being worth only 1.1 to 3.4 alone; the advantage shrinks under finite shots and reverses on two rows; and the comparison between the two best-arm variants is unresolved on the largest system.

Here g_i is the derivative of the energy with respect to the variational parameter of generator G_i at $\theta_i = 0$, in Hartree per radian; G_i is anti-Hermitian, so g_i is real. The generator pool and its labelling convention are fixed in Sec. 6.

2 Fixed method names

The phrase “naive/no-sharing” is retired as a *method* name. A construction that shares nothing does appear below, as the fourth corner of the two-by-two that the three methods span; it is called the *no-sharing baseline* throughout, it is a diagnostic rather than a project method, and it is defined in Sec. 11. App. B fixes the naming convention for code and files.

Table 1: Fixed method names.

Name	Full name	Definition
M1: FC-UG	Fully Commuting Universal Groups	Build one global parent FC grouping for the union of all Pauli products appearing in all ADAPT gradient commutators. Estimate the full gradient vector.
M2: BAI-FC-IG	BAI with Fully Commuting Individual Gradients	Treat each generator gradient as a BAI arm. For each active gradient, use an FC grouping of that individual commutator. No cross-gradient universal sharing is used.
M3: BAI-FC-UG	BAI with Fully Commuting Universal Groups	Use the same fixed parent FC universal groups as M1, but apply BAI elimination and sample only parent contexts needed to distinguish surviving generators.
<i>No-sharing baseline</i>	—	<i>Diagnostic, not a method.</i> Neither mechanism: each generator is grouped separately, as in M2, and every generator is resolved to the common radius used by M1 (Sec. 11).

3 Object lineage

The qubit Hamiltonian is

$$H = \sum_p h_p P_p, \quad (2)$$

where P_p are Pauli products after Jordan–Wigner mapping. For each generator G_i , define the gradient operator

$$C_i = [H, G_i]. \quad (3)$$

The ADAPT gradient is the expectation value of this operator,

$$g_i = \langle C_i \rangle_\psi. \quad (4)$$

Expanding all gradient operators in one shared Pauli-coordinate system gives

$$C_i = \sum_{\ell} A_{i\ell} R_{\ell}, \quad \mu_{\ell} = \langle R_{\ell} \rangle_{\psi}, \quad \mathbf{g} = A\boldsymbol{\mu}. \quad (5)$$

The parent universal Pauli support at this ADAPT step is

$$\mathcal{B}_0 = \bigcup_i \text{supp}(C_i). \quad (6)$$

M1 and M3 use $\mathcal{B}_0$ as a shared measurement object. M2 does not; it groups each C_i separately.

4 The three algorithms

4.1 M1: FC-UG

M1 estimates the full gradient vector. It forms one global FC grouping of the parent support,

$$\mathcal{B}_0 \longrightarrow \mathcal{M}_1, \dots, \mathcal{M}_K, \quad (7)$$

where each $\mathcal{M}_{\alpha}$ is a fully commuting measurement context. Measurements of these contexts estimate the shared Pauli expectations μ_{ℓ} and reconstruct every gradient using Eq. (5).

The implementation uses deterministic greedy FC grouping of $\mathcal{B}_0$ (Sec. 6). This is a global fully commuting all-gradient baseline in the spirit of the grouping strategies of Refs. [3, 2], in that it shares measurement contexts across the whole ADAPT gradient pool; it is not claimed to reproduce the specialised constructions of either. [AUTHOR: state what M2 and M3 add beyond Refs. [4, 3], and cite Refs. [1, 5] where they are used.]

4.2 M2: BAI-FC-IG

M2 changes the statistical target from full-vector estimation to winner identification. Each candidate gradient is an arm in a best-arm-identification problem. Its own commutator support is grouped separately,

$$\text{supp}(C_i) \longrightarrow \mathcal{M}_{i,1}, \dots, \mathcal{M}_{i,K_i}. \quad (8)$$

Throughout, t indexes the BAI round and r denotes a confidence radius; the two are never the same symbol. BAI maintains confidence intervals for active gradients and eliminates a generator i at round t if

$$|\widehat{g}_i| + r_i < \max_{j \in \mathcal{A}_t} (|\widehat{g}_j| - r_j), \quad (9)$$

where $\mathcal{A}_t$ is the active generator set and $r_i = z_{\delta} \epsilon_i$ is the current confidence radius of arm i , with ϵ_i its one-standard-error target and z_{δ} the confidence factor of Eq. (20). All methods in this report use a single common radius $r_i = r$ across active arms; the rule is written with per-arm radii because the covariance-aware variant of Sec. 15 does not. M2 does not exploit universal cross-gradient sharing.

Implementation status. M2 is now implemented as its own module. Each run writes the individual-gradient FC grouping $\mathcal{M}_{i,k}$ for every generator, the oracle fragment standard deviation of every fragment, the per-arm shot cost, and the full BAI elimination history with confidence radii and active sets. The M2 column is therefore auditable; it remains an oracle planning diagnostic rather than a finite-shot result.

4.3 M3: BAI-FC-UG

M3 combines the mechanisms of M1 and M2. It starts with the same fixed parent universal FC contexts as M1,

$$\mathcal{M}_1, \dots, \mathcal{M}_K, \quad (10)$$

but BAI eliminates generator candidates,

$$\mathcal{A}_0 \supset \mathcal{A}_1 \supset \mathcal{A}_2 \supset \dots. \quad (11)$$

At BAI round t , the active Pauli support is

$$\mathcal{B}_t = \bigcup_{i \in \mathcal{A}_t} \text{supp}(C_i), \quad \mathcal{B}_t \subseteq \mathcal{B}_0. \quad (12)$$

The active measurement contexts are restricted parent contexts,

$$\mathcal{M}_\alpha^{(t)} = \mathcal{M}_\alpha \cap \mathcal{B}_t. \quad (13)$$

The parent contexts are not rebuilt during BAI. This nested construction keeps previous measurements compatible while allowing the algorithm to stop sampling information that only affects eliminated generators.

How M3’s cost is aggregated. Shots taken in a context are reusable, so a context is charged once, at the largest number of shots any round ever demanded of it. Writing $n_\alpha^{(t)}$ for the allocation of Eq. (21) solved over the active set $\mathcal{A}_t$ at the radius of round t , the reported M3 cost is

$$M_{\text{M3}} = \sum_{\alpha} \max_t n_\alpha^{(t)}. \quad (14)$$

Each round’s allocation is solved independently, without reference to shots already committed; the maximum is taken afterwards. This is the accounting a real run would get for free by keeping its data, and it is why the trajectory is fully determined by the finitely many rounds at which the active set changes.

5 Oracle shot proxy

All numbers in Secs. 10–11 are oracle planning estimates. The state $|\psi\rangle$ is known exactly in the simulation, so the exact gradient values and exact fragment variances are available. No empirical finite-shot variance learning is included. *Oracle* and *noiseless* are used here as synonyms for this one property: the planner is given the exact g_i and $\sigma_{i\alpha}$. They do not refer to a noiseless quantum device, and they say nothing about state-preparation error, which is excluded from the model altogether (Sec. 7).

Sorted order and the gap. Order the pool by decreasing absolute gradient, $|g_{(1)}| \geq |g_{(2)}| \geq \dots$, so that (1) denotes the winner and (2) the runner-up. The parenthesised index is the rank, never a generator index. The top gap and the deficit of generator i are

$$\text{gap} = |g_{(1)}| - |g_{(2)}|, \quad \Delta_i = |g_{(1)}| - |g_i| \geq 0, \quad (15)$$

so $\Delta_{(2)} = \text{gap}$ and $\Delta_{(1)} = 0$.

Shots and context-shots. One *shot* is one state preparation followed by one measurement of one measurement context: a single circuit execution. A *context-shot* is the same thing counted against the context it was spent in; the two terms are interchangeable in this report, and every cost quoted is a total over contexts, $\sum_{\alpha} n_{\alpha}$. The metric counts shots only. It does not count the depth of the diagonalising circuits, the number of *distinct* measurement settings a method requires, or any classical post-processing;

Sec. 7 gives the distinct-setting counts, which differ between the methods by a factor of several and would have to be weighed separately in a device-level comparison.

For one gradient operator written as a sum of independently measured FC fragments,

$$C_i = \sum_{\alpha=1}^{K_i} F_{i\alpha}, \quad (16)$$

with oracle fragment variances

$$\sigma_{i\alpha}^2 = \text{Var}_\psi(F_{i\alpha}) = \langle F_{i\alpha}^2 \rangle_\psi - \langle F_{i\alpha} \rangle_\psi^2, \quad (17)$$

the optimal independent-fragment allocation gives the approximate total number of shots needed to estimate g_i with one-standard-error target ϵ_i as

$$M_i(\epsilon_i) \approx \frac{\left(\sum_{\alpha=1}^{K_i} \sigma_{i\alpha}\right)^2}{\epsilon_i^2}. \quad (18)$$

Precision target and confidence radius. ϵ_i is a one-standard-error target on $\hat{g}_i$; a *confidence radius* is the half-width of the corresponding interval,

$$r_i = z_\delta \epsilon_i, \quad (19)$$

so a method asked for radius r must reach $\epsilon = r/z_\delta$, and ϵ_i in Eq. (18) is replaced by r_i/z_δ . The confidence factor is the two-sided normal quantile with a Bonferroni correction over the whole pool,

$$z_\delta = \Phi^{-1}\left(1 - \frac{\delta}{2N}\right), \quad N = |\{G_i\}|, \quad (20)$$

with N the *full* pool size — 26 for H_4 , 92 for LiH , 140 for H_2O — not the number of non-zero gradients and not the current active set. No correction is taken over rounds. At $\delta = 0.05$ this gives $z_\delta = 3.10$ for H_4 , 3.46 for LiH and 3.57 for H_2O . The same z_δ is applied to all four constructions, so it cancels from every planning-bound ratio (Sec. 13). [AUTHOR: justify $\delta = 0.05$ and the Bonferroni correction.] Sec. 12.5 shows that this construction does *not* control the realised selection-error rate, so it should be read as a convention fixed identically across the methods rather than as a guarantee. For BAI the tolerance for an arm is set by the current separation between its estimated absolute gradient and the best surviving one.

Eq. (18) is applied separately inside each gradient arm, because M2 measures each arm in its own contexts. M1 and M3 instead share contexts between generators, so the allocation over contexts is not separable and is obtained by solving the convex programme

$$\min_{n_\alpha \geq 0} \sum_{\alpha} n_\alpha \quad \text{subject to} \quad \text{Var}(\hat{g}_i) = \sum_{\alpha} \frac{\sigma_{i\alpha}^2}{n_\alpha} \leq \epsilon^2 \quad \text{for every active } i, \quad (21)$$

where n_α is the number of shots taken in context α and $\text{Var}(\hat{g}_i) = \sum_{\alpha} \sigma_{i\alpha}^2 / n_\alpha$ is the *reconstruction variance* of gradient i under that allocation: the variance of the estimate obtained by reading $\hat{g}_i$ off the shared contexts through Eq. (5). The quantity $\sum_{\alpha} \sigma_{i\alpha}^2$ that appears repeatedly below is its value at one shot per context, and serves as a scale-free measure of how expensive generator i is to reconstruct from the parent grouping. Its stationarity condition is $n_\alpha = \sqrt{\sum_i \lambda_i \sigma_{i\alpha}^2}$ for multipliers $\lambda_i \geq 0$, which is solved by a damped multiplicative fixed-point iteration and then rescaled so that the tightest constraint holds exactly. The iteration uses multiplicative damping 0.5, at most 500 iterations, and stops when the least-loaded constraint is within 10^{-9} of the tightest; the final rescaling makes the returned allocation feasible

whatever the iteration converged to, so the stopping rule affects how close to optimal the answer is and never its validity. For a single gradient it reduces exactly to Eq. (18), and it agrees with a general-purpose convex solver on the small instances tested (Sec. 19). A previous envelope heuristic, which spread shots in proportion to $\max_i \sigma_{i\alpha}$, overestimated the M1 and M3 costs by up to a factor of eight and inverted the M2/M3 comparison; it has been replaced.

Shot counts are not rounded. n_α is left real-valued and no per-context minimum is imposed; the per-arm closed form of Eq. (28) used by M2 and the baseline is rounded up to an integer per arm. Imposing a floor of one shot per context would add at most K shots to a total, which is at most 0.2% of any entry in Table 6 — the worst case is the smallest total, M3 on H₄ side 2.0 CISD, where 53 contexts sit under 33,432 shots. It would change the digits quoted but not any ratio.

The BAI methods are evaluated in the schedule-free limit. As the common confidence radius r decreases continuously, generator i leaves the active set the moment $2r$ falls below its absolute-gradient deficit Δ_i of Eq. (15), so it is resolved to exactly $\Delta_i/2$; the leader survives to gap/2. Because measurements are reusable, an arm that leaves at radius r has paid the cost of being resolved to r and nothing more, and a shared context has paid the largest requirement it ever saw. The active set changes only at the finitely many distinct values of Δ_i , so the whole trajectory is determined by evaluating the allocation at those radii. For M2 this collapses to the closed form of Eq. (28). This replaces the geometric radius schedule of the previous revision, whose shrink factor was a free parameter and which, being coarse, overshoot the elimination points; Sec. 13 quantifies the difference.

Because these formulae depend on the exact grouping and fragment variances, every production run saves the FC groups, the fragment variances and the BAI histories used to generate each table entry.

6 Computational details

6.1 Molecular input

All cases use the minimal STO-3G basis, charge 0, and a closed-shell singlet restricted Hartree–Fock reference, with no frozen core and no qubit tapering. Table 2 gives the geometries in full. Water is placed in the xz plane with the oxygen at the origin and the bond angle held at 104.52° while $R(\text{OH})$ is varied, so the stretch is symmetric. H₄ is a planar square in the xy plane. [AUTHOR: state why these systems, geometries and states were chosen.]

Table 2: Cartesian geometries in Å. Basis STO-3G, charge 0, spin singlet, no frozen core, no tapering.

Case	Atom	Coordinates (x, y, z) / Å	Spatial orbitals
LiH, $R = 3.0$	Li	(0, 0, 0)	6
	H	(0, 0, 3.0)	
H ₄ square, side s	H	(0, 0, 0), (s , 0, 0), (0, s , 0), (s , s , 0) $s = 1.0$ and $s = 2.0$	4
H ₂ O, $R(\text{OH}) = R$	O	(0, 0, 0)	7
	H	$(\pm R \sin \frac{\theta}{2}, 0, R \cos \frac{\theta}{2})$, $\theta = 104.52^\circ$ $R = 0.9572$ and $R = 1.75$	

Two states are used per geometry. **HF** is the closed-shell Hartree–Fock determinant. **CISD** (configuration interaction with singles and doubles) is the lowest eigenvector of the qubit Hamiltonian restricted to the determinants within double excitation of that reference, in the target (N_α, N_β) sector, normalised to unity and embedded in the full 2^n -dimensional space. The CISD state is used as a correlated proxy for a later ADAPT state; it is not a saved ADAPT trajectory snapshot, and no evidence is offered here that its gradient spectrum resembles one. [AUTHOR: justify the CISD proxy.]

6.2 Generator pool, conventions and labels

Spin orbitals are interleaved: spatial orbital p gives spin orbitals $2p$ (α) and $2p + 1$ (β), zero-based. The computational basis is indexed big-endian, matching the sparse-operator convention of the linear-algebra backend. The Jordan–Wigner transform is used throughout.

The generator universe is the standard spin-orbital UCCSD pool. Each generator is anti-Hermitian,

$$G_i = \mathcal{JW}(\tau_i - \tau_i^\dagger), \quad \tau_i^S = a_a^\dagger a_i, \quad \tau_i^D = a_a^\dagger a_b^\dagger a_j a_i, \quad (22)$$

so that $g_i = \langle [H, G_i] \rangle_\psi$ of Eq. (1) is real and equals $\partial E / \partial \theta_i$ at $\theta_i = 0$ for the parameterisation $e^{\theta_i G_i}$. Singles run over occupied–virtual pairs of the same spin; doubles conserve N_α and N_β separately. The pool therefore conserves electron number and S_z but not generally S^2 . Ordering is deterministic: all singles in ascending (i, a) , then all doubles in ascending (i, j, a, b) . The labels used in the tables are exactly this: **S** $i \rightarrow a$ for a single and **D** $i, j \rightarrow a, b$ for a double, with zero-based interleaved spin-orbital indices. The resulting pool sizes are 26, 92 and 140 for H_4 , LiH and H_2O . [AUTHOR: justify this pool against qubit or spin-adapted alternatives.]

6.3 Fully commuting grouping routine

FC group counts are routine-dependent, and two rows are comparable only if produced by the same deterministic routine with the same Pauli ordering, tie-breaking rule and seed. Every row in the main tables uses one routine: deterministic greedy first-fit grouping, with Pauli strings sorted by decreasing weight (the number of non-identity factors) and ties broken by ascending lexicographic order of the string read left to right from qubit 0, under the letter order $I < X < Y < Z$, with each candidate joining the lowest-numbered existing group that accepts it and opening a new group otherwise. No random seed is used. Sec. 14 varies this convention deliberately and reports the spread. The counts differ from the previous revision, whose H_4 and H_2O rows used a different partitioning routine and were not comparable with the LiH row. [AUTHOR: justify fully commuting grouping against qubit-wise commuting and other schemes.]

6.4 Validation gate

Every case is checked before any gradient is computed: the Jordan–Wigner (JW) Hamiltonian must reproduce the PySCF restricted Hartree–Fock (RHF) energy on the Hartree–Fock (HF) determinant and the PySCF full configuration interaction (FCI) energy as its lowest eigenvalue in the (N_α, N_β) sector, both to 10^{-8} Ha. All nine cases pass, with a maximum observed error of 4×10^{-15} Ha. A case that fails the gate aborts rather than producing numbers.

6.5 Non-zero gradient threshold

Gradients that vanish by symmetry appear at the 10^{-8} numerical noise floor of the integral and SCF pipeline, while the smallest physically non-zero gradient in these cases is of order 10^{-3} . The counts below use a threshold of 10^{-6} , which lies on the plateau between the two. The threshold is not innocuous at tighter values: for LiH the count is 26 at 10^{-6} and 34 at 10^{-9} , and every run records the count at 10^{-4} , 10^{-6} , 10^{-8} and 10^{-9} so the choice stays visible. **The threshold is used for reporting only.** No method filters the pool by it: all four constructions carry every one of the N generators as an arm, and N in Eq. (20) is the full pool size. The one place the distinction is used analytically is the diagnostic in Sec. 10.1, which asks what M1 would cost if it were given the non-zero set for free.

7 What the model assumes and excludes

Statistical assumptions, in one place. (i) Estimators are Gaussian; the normal limit is assumed, not tested against shot-level sampling of measurement outcomes. (ii) Distinct measurement contexts are sampled independently. (iii) Within a context, the planning bounds and the main finite-shot tables treat different generators’ fragments as independent; this is exact for M2 and the baseline, which give every generator its own shots, and an approximation for M1 and M3. Sec. 15 removes it and reports what changes. (iv) State preparation is exact and noiseless, and there is no gate, readout or decoherence error anywhere in the model. (v) The union bound behind Eq. (20) is taken over the N pool members only; it is not taken over the sequence of elimination rounds, which in the exact limit is a continuum. (vi) Fragment variances are the exact $\text{Var}_\psi(F_{i\alpha})$, so no cost is charged for learning them.

“System size” means the pool. The phrase is used below to order the four systems, and the orderings induced by the plausible measures do not coincide, so it is fixed here: *system size* means the generator pool size N , which orders the systems H_4 (26) < LiH (92) < H_2O (140). On these four systems that ordering happens to agree with qubit count (8, 12, 14) and with $|\mathcal{B}_0|$ (1,276; 30,196; 81,566), but not with electron count, where LiH and H_2O both exceed H_4 in the other direction. Because only four systems are present and geometry and state vary with them, no claim in this report about a trend with system size is established as such; the phrase is a label for the ordering, not a scaling result.

What the cost metric leaves out. Every number counts shots. It does not count the depth or two-qubit gate count of the circuit that diagonalises a context, the cost of compiling or calibrating distinct measurement settings, or classical post-processing. The number of distinct settings differs sharply between the methods, and in the direction that disfavors the individually grouped ones.

Table 3: Distinct measurement settings each construction requires, a cost component the shot metric does not price. M1 and M3 share one parent grouping; M2 and the baseline group every generator separately, so their settings are the sum over arms and are not reused between arms.

Case	M1, M3: K	M2, baseline: $\sum_i K_i$	Largest single arm: $\max_i K_i$
LiH , $R = 3.0 \text{ \AA}$	980	5,744	101
H_4 square, side 1.0 and 2.0 $\text{\AA}$	53	183	10
H_2O eq., $R(\text{OH}) = 0.9572 \text{ \AA}$	2,366	13,235	159
H_2O stretch, $R(\text{OH}) = 1.75 \text{ \AA}$	2,356	14,263	162

M2 and the baseline need 3.5 to 6.1 times as many distinct settings as M1 and M3 here. Nothing in this report prices that, and a metric that did could move the M2-versus-M3 comparison, which on shots alone is already within a factor of two on most rows. The counts are structural properties of the groupings, not measured circuit costs.

8 Observable structure for all systems

Table 4: Global universal support used by M1: FC-UG and M3: BAI-FC-UG. The FC contexts are global parent contexts for $\mathcal{B}_0$, not per-gradient groupings, and all rows use the single greedy routine fixed in Sec. 6. Non-zero gradient counts use the 10^{-6} threshold. Distances in Å.

Case	State	Qubits	Pool	Universal terms	FC-UG contexts	Nonzero g_i
LiH, $R=3.0$	HF	12	92	30,196	980	26
H ₄ , side 1.0	HF	8	26	1,276	53	10
H ₄ , side 1.0	CISD	8	26	1,276	53	10
H ₄ , side 2.0	HF	8	26	1,276	53	10
H ₄ , side 2.0	CISD	8	26	1,276	53	10
H ₂ O eq., $R=0.9572$	HF	14	140	81,566	2,366	40
H ₂ O eq., $R=0.9572$	CISD	14	140	81,566	2,366	48
H ₂ O str., $R=1.75$	HF	14	140	81,566	2,356	40
H ₂ O str., $R=1.75$	CISD	14	140	81,566	2,356	48

The universal supports reproduce the earlier pilot data exactly (30,196 for LiH, 1,276 for H₄, 81,566 for H₂O), as do the gradient values and gaps in Table 5 to six or more digits. The FC context counts differ from the pilot’s because they now all come from one routine.

Why the two H₂O geometries have the same support size but different context counts.

The grouping routine is deterministic in the set of Pauli strings alone, so equal supports would have to give equal counts; 2,366 against 2,356 therefore means the supports are not equal. They are not. Both contain 81,566 products, and $\sum_i |\text{supp}(C_i)|$ is 314,720 in both, but their intersection is only 78,113: each geometry has 3,453 Pauli products the other lacks, a symmetric difference of 6,906, or 8.5% of each. The differing products are not numerical dust — their largest commutator coefficients reach 0.09 and 0.11, and their medians, 2.6×10^{-3} and 3.7×10^{-3} , are within a factor of two of the median over the shared products.

The cause is an orbital reordering. Both geometries have the same point group and the same multiset of orbital symmetries, but the Hartree–Fock orbital energies cross between them: the canonical order is $(A_1, A_1, B_2, A_1, B_1, A_1, B_2)$ at $R(\text{OH}) = 0.9572$ Å and $(A_1, A_1, B_1, B_2, A_1, A_1, B_2)$ at 1.75 Å. The symmetry-allowed integrals are the same in number, which is why $|\mathcal{B}_0|$ and $\sum_i |\text{supp}(C_i)|$ agree exactly, but they sit at different orbital indices, so they map to different Pauli strings under Jordan–Wigner. The same reordering is why the top generator is D 4,5 $\rightarrow$ 12,13 at equilibrium and D 6,7 $\rightarrow$ 12,13 when stretched. The two geometries are therefore comparable as physics but their Pauli-level structures are not identical, and a difference of ten contexts between them carries no information.

9 Gradient-selection difficulty

Table 5: Top-gradient gaps and M2: BAI-FC-IG oracle shot costs. Smaller gaps make winner identification harder; the BAI cost scales as the inverse square of the relevant deficit.

Case	State	Top generator	$ g_{(1)} $	$ g_{(2)} $	Gap	M2: BAI-FC-IG
LiH, $R=3.0$	HF	D 2,3 $\rightarrow$ 10,11	0.243187	0.146799	0.096388	1,140,583
H ₄ , side 1.0	HF	D 2,3 $\rightarrow$ 6,7	0.295316	0.273776	0.021539	854,629
H ₄ , side 1.0	CISD	D 0,1 $\rightarrow$ 6,7	0.142467	0.024386	0.118081	102,806
H ₄ , side 2.0	HF	D 2,3 $\rightarrow$ 6,7	0.315036	0.298387	0.016649	422,746
H ₄ , side 2.0	CISD	D 0,1 $\rightarrow$ 6,7	0.098961	0.047102	0.051859	86,857
H ₂ O eq., $R=0.9572$	HF	D 4,5 $\rightarrow$ 12,13	0.304910	0.203820	0.101090	25,096,606
H ₂ O eq., $R=0.9572$	CISD	D 4,5 $\rightarrow$ 12,13	0.006925	0.004861	0.002064	46,457,815,707
H ₂ O str., $R=1.75$	HF	D 6,7 $\rightarrow$ 12,13	0.388493	0.366638	0.021856	17,423,631
H ₂ O str., $R=1.75$	CISD	D 6,7 $\rightarrow$ 12,13	0.106873	0.089917	0.016957	168,305,383

All four constructions return the true maximum-gradient generator in all nine cases. This is a consistency check rather than a result: in oracle mode the elimination rule reads exact gradients, so it cannot eliminate the true maximiser, and selection is correct by construction. The question of whether the maximiser is still found once the gradients are estimated is the subject of Sec. 12.

10 Complete three-method comparison

All three methods are compared at one criterion: the winner must be separated from the rest of the pool at family-wise confidence $\delta = 0.05$. M1 reaches that with a single common radius $\text{gap}/2$ over the whole pool; M2 and M3 reach it by elimination, resolving each generator only to $\Delta_i/2$. No free schedule or precision parameter remains in any column.

M1 is given the exact gap, and this favours it. A non-adaptive method cannot know gap before measuring anything, yet the target radius $\text{gap}/2$ assumed for M1 throughout is computed from the exact gradients. M1 is therefore charged for the loosest radius that happens to suffice, which is the most generous reading of a non-adaptive method available; a real deployment would have to guess a radius, and Table 22 shows that the M1/M2 ratio on one H_4 row moves from 0.66 to 168 as that guess varies by a factor of sixteen. The same oracle knowledge enters the good-enough rule of Sec. 12.6, whose tolerance is set to $\text{gap}/4$. The best-arm methods do not need it: their stopping point is discovered from the data. Every M1 column below should be read as *M1 at its best*, and the sizes of the M1/M2 and M1/M3 factors are conditional on that choice.

Table 6: Oracle context-shot comparison at the winner-identification criterion. $M1/M2 > 1$ means M2 is cheaper than M1; $M2/M3 > 1$ means M3 is cheaper than M2. Entries below unity are set in bold.

Case	State	M1: FC-UG	M2: BAI-FC-IG	M3: BAI-FC-UG	M1/M2	M2/M3	M1/M3
LiH, $R=3.0$	HF	3,563,904	1,140,583	638,423	3.12	1.79	5.58
H_4 , side 1.0	HF	2,244,862	854,629	566,951	2.63	1.51	3.96
H_4 , side 1.0	CISD	59,642	102,806	53,727	0.58	1.91	1.11
H_4 , side 2.0	HF	598,102	422,746	220,389	1.41	1.92	2.71
H_4 , side 2.0	CISD	58,068	86,857	33,432	0.67	2.60	1.74
H_2O eq., $R=0.9572$	HF	172,147,028	25,096,606	22,130,852	6.86	1.13	7.78
H_2O eq., $R=0.9572$	CISD	411,524,732,152	46,457,815,707	44,971,732,317	8.86	1.03	9.15
H_2O str., $R=1.75$	HF	3,864,803,902	17,423,631	20,163,542	221.81	0.86	191.67
H_2O str., $R=1.75$	CISD	5,715,348,735	168,305,383	165,843,674	33.96	1.01	34.46

M3 is cheaper than M1 in every row. The intuition is that M3 solves the same allocation problem as M1 over a shrinking subset of the constraints, at radii never tighter than M1's, so each of its rounds costs no more than M1 does; but the reported M3 cost is the *sum over contexts of the per-context maximum* across rounds, Eq. (14), and a sum of maxima of separately optimised allocations is not obviously bounded by the optimum of the full problem. **No proof is offered here.** The ordering is checked numerically for every reported case, and holds; it should be read as an observation with a plausible mechanism, not a theorem. Sec. 19 records this as an open item for adversarial testing. M3 is cheaper than M2 in eight rows of nine, and M2 is cheaper than M1 in seven rows of nine; both statements are subject to the routine-dependence of Sec. 14.

For reference, M1 evaluated at the harder full-vector target $\text{gap}/4$ used by the previous revision costs exactly four times these values, since the cost scales as the inverse square of the radius.

10.1 What sets the M1 cost

M1 must reach its common radius for every generator simultaneously, so its cost is set by the pool member with the largest reconstruction variance $\sum_{\alpha} \sigma_{i\alpha}^2$ — the *binding generator*, in the sense that its constraint in Eq. (21) is the one the final rescaling makes tight. Table 7 identifies it in every case.

Table 7: The generator whose constraint binds M1, against the winner. $\sum_{\alpha} \sigma^2$ is the reconstruction variance at one shot per parent context. The last column is the factor by which M1’s cost falls if it is handed the set of non-zero-gradient generators for free — information no non-adaptive method has.

Case	Binding generator	$ g $ of it	$\sum_{\alpha} \sigma^2$	same, winner	M1 all / M1 non-zero
LiH, $R=3.0$, HF	D 0,1 $\rightarrow$ 4,9	0	11.76	0.072	2.28
H ₄ , side 1.0, HF	D 0,1 $\rightarrow$ 4,7	0	2.713	0.447	1.71
H ₄ , side 1.0, CISD	D 0,1 $\rightarrow$ 6,7	1.4×10^{-1}	1.741	1.741	1.56
H ₄ , side 2.0, HF	D 0,1 $\rightarrow$ 4,7	0	0.218	0.067	1.83
H ₄ , side 2.0, CISD	S 1 $\rightarrow$ 7	1.9×10^{-31}	0.144	0.122	1.93
H ₂ O eq., HF	S 0 $\rightarrow$ 10	7.1×10^{-14}	537.4	1.317	2.20
H ₂ O eq., CISD	S 0 $\rightarrow$ 10	1.6×10^{-5}	530.8	1.284	2.01
H ₂ O str., HF	S 0 $\rightarrow$ 10	1.8×10^{-8}	531.6	0.592	2.35
H ₂ O str., CISD	S 0 $\rightarrow$ 10	2.0×10^{-4}	455.9	0.565	2.02

In six of the nine cases the binding generator has a gradient below the 10^{-6} threshold, and its reconstruction variance dwarfs the winner’s — by a factor of 164 on LiH and of 400 to 900 on H₂O. In the remaining three it carries a small but non-zero gradient, and on H₄ side 1.0 with the CISD state it is the winner itself.

This is not a defect of the comparison: M1 is non-adaptive, so it must bound every generator tightly enough to rule it out, and paying for high-variance operators that turn out not to matter is precisely the inefficiency elimination removes. Nor is it the whole story. Restricting M1 to the non-zero-gradient generators — the last column of Table 7 — cuts its cost by only 1.6 to 2.4, because the rest also carry substantial variance. The cost is spread over the pool rather than concentrated in one irrelevant operator. This is also a rough guide to how much Part III symmetry filtering could do for M1, although not a bound: filtering and tapering transform the operators as well as dropping generators, so the two are not the same operation.

10.2 Where the cost actually sits: breadth and depth

The shot cost of a BAI method is not concentrated in the final elimination, as a naive reading of the $1/r^2$ scaling suggests. It splits into two terms, and which one dominates changes with the system.

One definition, applied the same way to both methods. Let $r_0 = \frac{1}{2} \max_i \Delta_i$ be the coarsest radius of the trajectory, the radius at which the first generator is eliminated and at which every generator is still active. *Breadth* is the cost of that first sweep,

$$M^{\text{breadth}} = \text{cost of resolving every generator to } r_0, \quad M^{\text{depth}} = M^{\text{total}} - M^{\text{breadth}}, \quad (23)$$

and the *breadth share* is $M^{\text{breadth}}/M^{\text{total}}$. For M2 the cost is separable, so $M^{\text{breadth}} = \sum_i (z_{\delta} S_i^{\text{own}}/r_0)^2$, writing $S_i^{\text{own}} = \sum_{\alpha} \sigma_{i\alpha}$ over generator i ’s own contexts and S_i^{par} for the same sum over the parent contexts. For M3 it is the total of the allocation at the first breakpoint, before any elimination. Because shots are reusable, breadth is a cost neither method can avoid: it is the price of learning which generators are worth pursuing.

Table 8: Breadth share, Eq. (23): the fraction of each method’s total that is already committed at the coarsest radius, when every generator is still active. The remainder is the depth term. Both columns use the same definition; M3’s is the cumulative allocation at the first breakpoint of its elimination history.

Case	M2 breadth share	M3 breadth share	$\max_i \Delta_i/\text{gap}$
H ₄ , side 1.0, HF	3.0%	2.1%	13.7
H ₄ , side 1.0, CISD	87.1%	76.3%	1.2
H ₄ , side 2.0, HF	1.3%	0.8%	18.9
H ₄ , side 2.0, CISD	57.1%	47.7%	1.9
LiH, $R=3.0$, HF	93.9%	87.7%	2.5
H ₂ O eq., HF	89.4%	85.5%	3.0
H ₂ O eq., CISD	92.8%	81.3%	3.4
H ₂ O str., HF	80.1%	60.7%	17.8
H ₂ O str., CISD	94.5%	86.8%	6.3

The two HF H₄ rows are strongly depth-dominated: only 1 to 3% of the cost is committed before the first elimination, and essentially all of it goes into separating the leading pair. LiH, both H₂O geometries and the two H₄ CISD rows are breadth-dominated, with 57 to 94% of the cost already spent when every generator is still in play. What distinguishes them is $\max_i \Delta_i/\text{gap}$, the dynamic range the schedule has to cover: 13.7 and 18.9 on the two depth-dominated H₄ HF rows against 1.2 to 3.4 on H₄ CISD, LiH and H₂O equilibrium. Where the leading pair is nearly tied relative to the spread of the field, depth dominates; where the deficits are spread more evenly, breadth does. H₂O stretch, at 17.8 and 6.3, does not fit that reading, so the correlation is suggestive rather than established.

The two terms reward sharing very differently. In the breadth term a shared context serves the entire pool, 92 or 140 generators at once, and M3’s advantage over M2 is correspondingly large. In the depth term it serves only the generators that survive to the final radius. M3’s breadth share is below M2’s in every one of the nine rows, which is what sharing is supposed to do: it makes the coarse sweep cheaper relative to the rest.

10.3 What decides the depth term, and why the M3 advantage may not survive

Two quantities decide the depth term. The first is the *fragmentation penalty*. Generator i ’s own FC grouping is $\mathcal{M}_{i,1}, \dots, \mathcal{M}_{i,K_i}$ as in Sec. 4; write $F_{i\alpha}$ for the part of C_i that lands in parent context $\mathcal{M}_\alpha$, so that the parent grouping splits C_i into K_i^{parent} pieces instead of $K_i^{\text{own}} \equiv K_i$. Then

$$f_i = \left(\frac{\sum_\alpha \sigma_{i\alpha} \text{ over the parent contexts}}{\sum_k \sigma_{ik} \text{ over its own contexts } \mathcal{M}_{i,k}} \right)^2, \quad (24)$$

is the factor by which the shared grouping makes that one gradient more expensive to resolve on its own. It is above unity in every case measured here, and the reason is that Eq. (18) depends on $(\sum_\alpha \sigma_{i\alpha})^2$ and not on $\sum_\alpha \sigma_{i\alpha}^2$, so splitting an operator into more pieces increases the sum of the pieces’ standard deviations. This is not a theorem: the parent grouping is not a refinement of the generator’s own grouping — a parent context can merge Paulis that the own grouping separates — so $f_i \geq 1$ is an observation, supported by K^{parent} exceeding K^{own} by factors of 2 to 16 in Table 9. The second quantity is

$$|\mathcal{A}| = |\{i : \Delta_i \leq \text{gap}\}|, \quad (25)$$

the number of generators still active at the final radius $\text{gap}/2$. By construction $|\mathcal{A}| \geq 2$, the leader and the runner-up.

In Table 9, K^{own} is the number of groups in the winner’s own FC grouping and depends only on $\text{supp}(C_i)$, hence not on the state. K^{parent} is the number of parent contexts in which the winner has a *non-zero* fragment standard deviation. It therefore does depend on the state, which is why it differs between the HF and CISD rows of H₂O (858 against 887, and 875 against 899) even though the winner and its

support are the same operator: a fragment whose variance vanishes in one state carries information in the other. Contexts where $\sigma_{i\alpha} = 0$ contribute nothing to Eq. (18) and are not counted.

Table 9: Fragmentation of the winning gradient against the number of finalists. K^{own} is the number of groups in the winner’s own FC grouping; K^{parent} is the number of parent contexts in which it has a non-zero fragment standard deviation, which is why it depends on the state. Both are defined in the text below.

Case	State	K^{own}	K^{parent}	f	$ \mathcal{A} $	M2/M3
H ₄ , side 1.0	HF	10	22	1.13	2	1.51
H ₄ , side 1.0	CISD	7	28	1.07	3	1.91
H ₄ , side 2.0	HF	10	22	1.19	3	1.92
H ₄ , side 2.0	CISD	7	28	1.22	2	2.60
LiH, $R=3.0$	HF	58	520	3.49	3	1.79
H ₂ O eq., $R=0.9572$	HF	57	858	2.54	2	1.13
H ₂ O eq., $R=0.9572$	CISD	57	887	2.44	2	1.03
H ₂ O str., $R=1.75$	HF	50	875	3.62	2	0.86
H ₂ O str., $R=1.75$	CISD	50	899	2.94	2	1.01

The scaling looks asymmetric. $|\mathcal{A}|$ is set by how many generators are nearly tied at the top, which has nothing to do with pool size and is two or three in every case here. The fragmentation penalty is larger on the larger systems, because a larger universal support spreads each individual gradient over more parent contexts: 22 on H₄, 520 on LiH, roughly 875 on H₂O. If that persisted, M3’s benefit in the depth term would have a ceiling while its cost in that term did not, and a crossover would follow.

This is a hypothesis, not a demonstration. f is not monotone in system size — LiH, the smaller system, gives 3.49 against 2.54 for H₂O at equilibrium — four systems cannot separate size from geometry and state, and f grows far more slowly than the raw context count (3.6 against 17.5 on H₂O stretch), because most parent contexts hold a negligible sliver of any one gradient. The mechanism is real but weak, and nothing here fixes where or whether a crossover occurs. Sec. 12 finds the opposite trend under finite shots; Sec. 12.4 reconciles them.

10.4 The finalists do not share Pauli terms

The premise of universal grouping is that gradient operators share Pauli products. Across the pool they do: each Pauli product in $\mathcal{B}_0$ appears in roughly four different gradient commutators.

Table 10: Support sharing. Average reuse is $\sum_i |\text{supp}(C_i)|/|\mathcal{B}_0|$. Finalist overlap is the Jaccard overlap $|\text{supp}(C_{(1)}) \cap \text{supp}(C_{(2)})|/|\text{supp}(C_{(1)}) \cup \text{supp}(C_{(2)})|$ between the supports of the two leading generators. Supports do not depend on the state, but the identity of the finalists does, so the two H₄ rows differ. The remaining four cases have not been evaluated.

Case	$ \mathcal{B}_0 $	$\sum_i \text{supp}(C_i) $	Average reuse	Finalist overlap	M2/M3
H ₄ , side 1.0, HF	1,276	4,148	3.25	0.0%	1.51
H ₄ , side 1.0, CISD	1,276	4,148	3.25	27.8%	1.91
LiH, $R=3.0$, HF	30,196	120,384	3.99	35.1%	1.79
H ₂ O eq., $R=0.9572$, HF	81,566	314,720	3.86	0.0%	1.13
H ₂ O str., $R=1.75$, HF	81,566	314,720	3.86	0.0%	0.86

The two columns tell opposite stories. Pool-wide reuse of about four is why M3’s breadth term is efficient. But in three of the five cases the two finalists share *no* Pauli products at all, so at the depth stage there is nothing to share and M3 pays the fragmentation penalty for nothing.

The overlap column does not order the M2/M3 column across systems. The two cases with substantial finalist overlap, LiH and H₄ CISD, are cases where M3 keeps a clear advantage, and the H₂O rows with none are where it disappears — but H₄ HF has zero overlap and still gives 1.51, well above both H₂O

rows, and its fragmentation penalty is six times smaller than LiH’s. Low finalist overlap is a plausible reason for M3 to lose its depth-stage advantage and is present where the advantage is smallest; five rows are not enough to make it *the* explanation.

10.5 An alternative: rebuild the grouping once the field narrows

If the fixed parent contexts are the problem at the depth stage, an obvious remedy is to abandon them once the active set is small and build a fresh fully commuting grouping over only the survivors’ union support, accepting that the earlier shots were taken in different circuit settings and do not transfer. This was tested directly. Phase one uses the parent contexts while more than a chosen number of generators remain active; phase two regroups from scratch, using the same greedy first-fit routine and ordering as the parent grouping, and is charged the full cost of reaching the final radii, ignoring the head start phase one has already provided, so the totals below are upper bounds. Switch points $|\mathcal{A}| \leq 2, 3, 4, 6, 10, 20$ were tested on each case and the best is reported; because the active set jumps in large steps, several thresholds often fire at the same breakpoint and give identical totals.

Table 11: Staged regrouping against the better of M2 and M3. The switch point is the active-set size at which the parent contexts are abandoned.

Case	Better of M2, M3	Best staged total	Switch at	Ratio
H ₄ , side 1.0, HF	566,951	502,858	$ \mathcal{A} \leq 10$	0.89
LiH, $R=3.0$, HF	638,423	638,423	never	1.00
H ₂ O str., $R=1.75$, HF	17,423,631	16,880,737	$ \mathcal{A} \leq 3$	0.97

Staged regrouping does beat both methods, but only by 3 to 11%, and the best switch point is not consistent: early on H₄, late on H₂O, and never on LiH, where the procedure degenerates to M3. The reason is visible in Table 8. Regrouping can only improve the depth term, and on the systems where M3 struggles the cost is dominated by breadth, which the remedy does not touch. On H₂O stretch it improves M3 by 16% and the better of the two methods by 3%.

A second observation from the same experiment bears on where regrouping should act. On H₄ side 1.0 HF the best switch point is the earliest one available: the very first elimination breakpoint, at which sixteen of the twenty-six generators leave at once and ten remain. Regrouping the survivors’ union support at that moment needs twelve contexts against the parent grouping’s fifty-three. The reason is that the union support collapses disproportionately, and it can be quantified exactly on this case: the sixteen zero-gradient generators account for 708 of the 1,276 Pauli products in $\mathcal{B}_0$, or 55.5%, and their supports are *disjoint* from those of the ten non-zero generators, whose union is the remaining 568. Dropping them therefore removes more than half of $\mathcal{B}_0$ at a stroke.

This is a single case, and the disjointness is a property of the H₄ square that should not be assumed elsewhere; the corresponding shares have not been measured on LiH or H₂O. Subject to that, it suggests that if regrouping is pursued, the leverage is not at the final pair but as soon as the first substantial group of candidates has been ruled out — which is also the point at which Part III symmetry filtering would act, without needing to measure anything first.

11 Attribution: the no-sharing baseline

The three methods are two binary choices, sharing and elimination, and they span only three of the four combinations. The fourth is a baseline that does neither: every generator is measured in its own fully commuting grouping, and every one of them is resolved to the same common radius used by M1. It is reported here as a diagnostic outside the method hierarchy, and it serves two purposes. It makes the two mechanisms separately attributable, and it is a sanity bound, since measuring every generator separately is always available as a fallback.

Table 12: Cost of the no-sharing baseline and the improvement each mechanism buys against it. Sharing only is baseline/M1, elimination only is baseline/M2, both is baseline/M3. The product column is what “both” would be if the two mechanisms acted independently, and the interaction is the ratio of the two.

Case	State	Baseline	Sharing only	Elimination only	Both	Product	Interaction
LiH, $R=3.0$	HF	6,819,980	1.91	5.98	10.68	11.44	0.93
H ₄ , side 1.0	HF	4,848,846	2.16	5.67	8.55	12.25	0.70
H ₄ , side 1.0	CISD	130,375	2.19	1.27	2.43	2.77	0.88
H ₄ , side 2.0	HF	2,002,941	3.35	4.74	9.09	15.87	0.57
H ₄ , side 2.0	CISD	180,604	3.11	2.08	5.40	6.47	0.84
H ₂ O eq., $R=0.9572$	HF	204,022,319	1.19	8.13	9.22	9.63	0.96
H ₂ O eq., $R=0.9572$	CISD	485,001,056,522	1.18	10.44	10.78	12.30	0.88
H ₂ O str., $R=1.75$	HF	4,407,619,149	1.14	252.97	218.59	288.50	0.76
H ₂ O str., $R=1.75$	CISD	6,317,143,181	1.11	37.53	38.09	41.49	0.92

Two things follow, and both bear on how the project should be described.

The mechanisms are substitutes, not complements. The interaction is below unity in every one of the nine cases, between 0.57 and 0.96. Applying both mechanisms buys less than the product of applying each alone. This is the quantitative form of the argument in Sec. 10.2: elimination removes precisely the situation that sharing exploits, namely many generators active at once, so the second mechanism applied always finds less left to gain.

Sharing contributes less than elimination, and less on the larger systems. On its own, universal grouping is worth a factor of 2.2 to 3.4 on H₄, 1.9 on LiH, and only 1.1 to 1.2 on H₂O. Elimination on its own is worth between 1.27 and 252.97: the low end is the H₄ side 1.0 CISD row, where the gap is wide and there is little for elimination to save, and the high end is the H₂O stretch HF row. The two ranges overlap, so “sharing contributes little” should be read as *relative to elimination* and not as an absolute statement — a factor of 3.4 is a real saving on its own terms. What the table does establish is the ordering of the two mechanisms and its direction: the sharing factor is smallest on the largest system in this set, on all four H₂O rows. Whether that is a size effect or a property of these particular molecules cannot be settled from four systems.

12 Finite-shot behaviour

Every number above is an oracle planning bound. The elimination rule reads exact gradients, so it cannot misfire, and the family-wise confidence is asserted rather than measured. This section replaces the exact gradients with estimates built from a finite number of shots and runs the same elimination rule on those.

Protocol. A real run does not know the deficits in advance and cannot jump to the exact breakpoints, so each trial follows a geometric radius schedule: the common radius starts at $\max_i |g_i|$, wide enough that no arm is eliminated for free, and is multiplied by $\gamma = 0.5$ each round. At each round the allocation of Eq. (21) is solved for the current active set and radius, and each context’s shot count is raised to that value if it is higher, never lowered; the estimate $\hat{g}_i$ is then redrawn from $\mathcal{N}(g_i, \sum_{\alpha} \sigma_{i\alpha}^2/n_{\alpha})$ at the accumulated n_{α} , which is the distribution of the estimator built from all shots taken so far. The elimination rule of Eq. (9) is then applied. A trial ends when one arm remains or the radius falls below $10^{-3} \max_i |g_i|$. Every case uses 200 trials at seed 0.

The reported inflation mixes two effects and this study does not separate them. The schedule is the same coarse $\gamma = 0.5$ that Sec. 13 shows overpaying at the planning level, by 1.6 to 3.8 across these cases with no noise at all. So an inflation of, say, 2.2 is not 2.2 worth of measurement noise: an unknown part of it is discretisation. The two cannot be read off against each other here, because the planning model with $\gamma = 0.5$ resolves the leader to gap/2 while a trial stops as soon as one arm remains, so the noiseless and noisy runs do not answer the same question. What is unaffected is the *comparison*

between M2 and M3, which use the same schedule, the same initial radius and the same stopping rule. Separating schedule from noise needs a designed experiment and is listed in Sec. 18.

The two deliberate approximations — the normal limit, and independent fluctuations per generator — are stated once with the rest in Sec. 7. The second is exact for the baseline and M2 and an approximation for M1 and M3; Sec. 15 removes it.

Columns. *Bound* is the oracle planning cost of Table 6; *Infl.* is the mean total shots over the 200 trials divided by it; *Correct* is the fraction of trials returning the true maximum-gradient generator. M2/M3 columns are ratios of the corresponding entries, standard errors combined in quadrature.

Table 13: Realised cost against the planning bound, 200 trials per row at seed 0, all uncertainties one standard error of the mean. Every row selected the true maximiser in all 200 trials, a Wilson 95% interval of $[0.981, 1]$; the *Correct* column is therefore constant and omitted. The non-adaptive baseline and M1 realise their bound exactly, having no schedule to overshoot, and are omitted.

Case	State	M2: BAI-FC-IG		M3: BAI-FC-UG		M2/M3	
		Bound	Infl.	Bound	Infl.	Bound	Realised
LiH, $R=3.0$	HF	1,140,583	3.16 ± 0.06	638,423	1.40 ± 0.03	1.79	4.03 ± 0.12
H ₄ , side 1.0	HF	854,629	2.21 ± 0.08	566,951	2.33 ± 0.10	1.51	1.43 ± 0.08
H ₄ , side 1.0	CISD	102,806	2.99 ± 0.05	53,727	2.74 ± 0.04	1.91	2.09 ± 0.04
H ₄ , side 2.0	HF	422,746	2.33 ± 0.05	220,389	2.48 ± 0.05	1.92	1.81 ± 0.05
H ₄ , side 2.0	CISD	86,857	2.77 ± 0.06	33,432	2.95 ± 0.09	2.60	2.43 ± 0.09
H ₂ O eq., $R=0.9572$	HF	25,096,606	3.02 ± 0.06	22,130,852	1.53 ± 0.04	1.13	2.24 ± 0.07
H ₂ O str., $R=1.75$	HF	17,423,631	2.75 ± 0.06	20,163,542	1.66 ± 0.04	0.86	1.43 ± 0.04

Realised means are Bound $\times$ Infl.; the largest is 7.6×10^7 shots for M2 on H₂O at equilibrium and the smallest 9.9×10^4 for M3 on H₄ side 2.0 CISD.

The two H₂O CISD rows are absent because they have not been simulated; they are the most expensive cases and were left out of this round rather than excluded for any reason bearing on the result (Sec. 18).

The scatter is not negligible and is the reason the errors are quoted. Repeated runs of the same configuration at different seeds move a realised total by 2 to 4%, which is comparable to some of the differences between adjacent rows and larger than the difference between the bound and realised M2/M3 ratios on the H₄ HF rows.

12.1 Finite shots reorder the comparisons with M1

M1 and the no-sharing baseline are non-adaptive: they solve one allocation at one radius and realise their planning bound exactly, with no schedule to overshoot. The best-arm methods do not. Comparing realised cost against M1’s bound therefore gives a different ranking from Table 6, and on several rows it reverses.

Table 14: M1 against the *realised* cost of the best-arm methods. M1’s bound is its realised cost. Entries below unity, where the best-arm method is more expensive than static all-gradient estimation, are set in bold.

Case	State	M1	M1 / M2 realised	M1 / M3 realised
LiH, $R=3.0$	HF	3,563,904	0.99	3.99
H ₄ , side 1.0	HF	2,244,862	1.19	1.70
H ₄ , side 1.0	CISD	59,642	0.19	0.40
H ₄ , side 2.0	HF	598,102	0.61	1.09
H ₄ , side 2.0	CISD	58,068	0.24	0.59
H ₂ O eq., $R=0.9572$	HF	172,147,028	2.27	5.07
H ₂ O str., $R=1.75$	HF	3,864,803,902	80.7	115.3

On the planning bounds M3 beats M1 on all seven of these rows and M2 on five. On realised cost M3 beats M1 on five and M2 on three. Both H_4 CISD rows invert for both methods, and LiH becomes a tie for M2. The pattern is that elimination pays where M1 is expensive — H_2O , where the factor stays between 2 and 115 — and loses where M1 is already cheap, which is exactly the regime the planning bounds identified in conclusion 4, but the finite-shot penalty moves two more rows across the line.

Some of this is the coarse schedule rather than noise, as noted above, and a deployment would tune the schedule. It nonetheless means the headline factors of 1.1 to 192 from Table 6 are the *bound-level* advantage of M3 over M1, and that the realised advantage is smaller everywhere and negative on two of seven rows.

12.2 The one inversion does not survive

On H_2O stretch with the Hartree–Fock reference, the planning bounds give $M2/M3 = 0.86$, the only row in which M3 is more expensive than M2. Under finite shots the same row gives 1.43 ± 0.04 , and M3 wins. The inversion is an artefact of comparing two bounds that inflate by different amounts, and the conclusion drawn from it in the previous revision is retracted in Sec. 16.

This does not establish the opposite verdict. Sec. 14 shows the same quantity moving over 0.74 to 1.30 on the bounds when the grouping routine is changed, and the finite-shot study was run under one routine only. The conclusion available here is the weaker one: the bound-level inversion is not robust, so it cannot be used to claim that M2 beats M3 on that row.

12.3 The two BAI methods inflate differently

The inflation is not a common factor. M2 inflates by 2.2 to 3.2 on every row. M3 inflates by 2.3 to 3.0 on all four H_4 rows, where it tracks M2 closely and the $M2/M3$ ratio is preserved to within about 10%, but by only 1.40 to 1.66 on LiH and both H_2O rows, where the $M2/M3$ ratio roughly doubles. The separation is far outside the quoted standard errors.

The natural explanation is overshoot. A real run does not know the deficits Δ_i in advance, so it discovers them by shrinking a common radius and overshooting each elimination point. M3’s breadth term is a single coarse sweep of the whole pool, a fixed cost that no schedule can overshoot, whereas every one of M2’s per-generator costs is discovered by overshooting. On that reading the more breadth-dominated the system, the less of M3’s cost is exposed to the schedule.

The data support the effect but not that explanation of it. If breadth-dominance were the mechanism, the two H_4 CISD rows should behave like LiH and H_2O : Table 8 puts their M2 breadth shares at 87.1% and 57.1%, higher than H_2O stretch’s 80.1% in one case. They do not. Both show M3 inflating as much as M2, exactly like the depth-dominated H_4 HF rows. What the seven rows actually separate on is the system, not the breadth share: the three large-pool cases behave one way and all four H_4 cases the other. Pool size, context count and breadth share are confounded across four systems, and the overshoot mechanism remains a hypothesis that a designed experiment — the same system at several pool sizes, or the same pool at several schedules — has not yet been run to test.

What survives without an explanation is the observation itself: on the three largest cases here, universal contexts help substantially more under finite shots than the planning bounds suggest.

12.4 Reconciling the two directions

Sec. 10.3 argues that M3’s advantage should *decay* as the system grows; this section finds it *growing*. They are not in contradiction, because they are statements about different quantities, and it is worth being explicit about which.

The fragmentation argument is about the *planning bound*, and within it about the depth term only: as the parent support grows, each individual gradient is split over more parent contexts, so resolving the

final pair through shared contexts costs more relative to resolving it through their own. That mechanism is real and is measured in Table 9. The finite-shot result is about *realised* cost under a schedule, which the bound does not model at all. The two therefore do not compete, and no mechanism is offered here for the second: the previous subsection records that the obvious one fails.

Two cautions. The realised M2/M3 ratios that drive the second trend are largest on H₂O (2.24 and 1.43), and conclusion 3 withdraws every M2-versus-M3 verdict on H₂O because the bound-level ratio moves across unity with the grouping routine; the finite-shot study was run under one routine, so those two entries inherit the same caveat and should not be used on their own. And “fading” in the conclusions refers only to the bound-level attribution of Sec. 11, where the sharing-only factor falls from 3.4 on H₄ to 1.1 on H₂O. Neither trend is established as a function of size: both rest on four systems in which size, geometry and state vary together.

12.5 The confidence convention is far stricter than δ , by an amount the data can only bound

No trial in any run selected the wrong generator. Sweeping the family-wise failure probability over nearly its whole range barely changes that.

How much stricter cannot be measured here. With 300 trials and zero errors, a Wilson 95% interval puts the true selection-error rate below 1.3% and no lower: the data are consistent with any rate from 0 to about 1%. The figure of 1.2×10^{-5} quoted below is the *analytic prediction* of Eq. (26), not a measurement, and resolving a rate that small would need on the order of 10^6 trials. What the experiment establishes is that the realised rate is at most 1.3% where $\delta = 0.05$ was asked for, about four times smaller, and that it barely moves when δ is swept over nearly its whole range. That is enough for the practical conclusion — δ is not the knob that controls the selection-error rate — and not enough to say by how much the convention over-delivers.

Table 15: Calibration sweep on H₄ side 1.0, HF, 300 trials per entry. The stated δ has almost no influence on the realised selection-error rate. Wilson 95% intervals on the correct-selection rate are given in brackets; at 300 trials none of these rows distinguishes a rate of 10^{-5} from one of 10^{-2} .

δ	z_δ	M2 correct [95% CI]	M3 correct [95% CI]	M2 shots	M3 shots
0.05	3.10	100.0% [98.7, 100]	100.0% [98.7, 100]	1,891,179	1,324,843
0.40	2.42	100.0% [98.7, 100]	100.0% [98.7, 100]	1,232,695	849,844
0.90	2.11	99.3% [97.6, 99.8]	99.7% [98.1, 99.9]	1,128,410	691,975
0.99	2.07	100.0% [98.7, 100]	99.7% [98.1, 99.9]	969,099	741,804

The reason is structural rather than a defect of the simulation. The procedure stops when the common radius reaches $\text{gap}/2$, at which point the standard error of each estimate is $\text{gap}/(2z_\delta)$. The leader’s margin over the runner-up is one gap, so in units of the standard deviation of their difference it is $2z_\delta/\sqrt{2} = z_\delta\sqrt{2}$, and the selection-error probability is approximately

$$P(\text{wrong selection}) \approx 2\bar{\Phi}(z_\delta\sqrt{2}), \quad (26)$$

where $\bar{\Phi}$ is the upper-tail standard normal function. This is 1.6×10^{-1} at $z_\delta = 1$, 4.7×10^{-3} at $z_\delta = 2$ and 1.2×10^{-5} at the $z_\delta = 3.1$ used in the tables. The displacement probability for a single competitor is $\bar{\Phi}(z_\delta\sqrt{2})$; the factor of two is a conservative allowance, not a derivation, covering the fact that the comparison is between absolute values and that more than one arm may be near the top. Dropping it would halve every predicted rate and change no conclusion. The Bonferroni term enters only through z_δ , and the stopping radius already supplies most of the protection.

The observed rates are consistent with Eq. (26) and inconsistent with δ , in the following restricted sense. At $\delta = 0.05$ and 0.40 the prediction is below 10^{-3} and the observation is 0/300, which excludes

δ itself but does not resolve the prediction. At $\delta = 0.90$ and 0.99 the prediction rises to 2.9×10^{-3} and 3.4×10^{-3} , and the observations, 1 to 2 errors in 300, have Wilson intervals of $[0.2, 2.4]\%$ and $[0.1, 1.9]\%$, which contain the prediction and exclude δ . The expression is therefore supported where it is testable and unresolved where it is not.

The practical consequence is that δ as used does not control the selection-error rate. Inverting Eq. (26) for a genuine 5% error suggests $z_\delta = 1.386$ against the 3.10 used in the tables, a factor of five in the planning bound. Simulation does not bear that out.

Table 16: Realised error and cost against the confidence factor, H_4 side 1.0, HF, 500 trials per entry. Brackets are Wilson 95% intervals on the realised error rate.

z_δ	Eq. (26)	M2 error [95% CI]	M3 error [95% CI]	M2 shots	M3 shots
1.39	0.049	12.2% [9.6, 15.4]	10.6% [8.2, 13.6]	1,586,040	989,854
1.80	0.011	1.8% [1.0, 3.4]	2.0% [1.1, 3.6]	1,113,341	819,684
2.10	0.003	0.4% [0.1, 1.5]	0.0% [0, 0.8]	1,211,688	685,567
3.10	1.2×10^{-5}	0.0% [0, 0.8]	0.0% [0, 0.8]	2,004,737	1,335,779

Eq. (26) accounts only for the final comparison and ignores generators wrongly eliminated earlier in the run, so it underestimates the error at small z_δ : it predicts 5% at $z_\delta = 1.39$ where the simulation gives ten to twelve. The cost is also not monotone. Below $z_\delta \approx 1.8$ it rises again, because bad eliminations prolong the run. There is an optimum near $z_\delta \approx 1.8$ to 2.1, worth about 1.7 in realised cost at an error rate whose 95% interval reaches 3.6% at worst — a real saving, but not the factor of five the bound suggested. Both conventions are selectable in the implementation; the tables in this report retain the Bonferroni one. The sweep was run on one case, H_4 side 1.0 with the Hartree–Fock reference, and the optimum has not been checked on any other, so $z_\delta \approx 1.8$ to 2.1 should be treated as a single-system observation rather than a recommended default.

12.6 Good-enough selection

A run does not have to identify the maximiser; it is often enough to return a generator that is not much worse. Fix a tolerance $\tau > 0$ — written τ to keep it distinct from the deficits Δ_i — and stop at the first round t at which every surviving candidate is certified within τ of the best,

$$\max_{j \in \mathcal{A}_t} (|\hat{g}_j| - r_j) \geq \max_{i \in \mathcal{A}_t} (|\hat{g}_i| + r_i) - \tau, \quad (27)$$

returning $\arg \max_{i \in \mathcal{A}_t} |\hat{g}_i|$. The run is counted a success if the returned generator’s true absolute gradient is within τ of the true maximum.

On H_4 side 1.0 HF at $\tau = \text{gap}/4 = 5.38 \times 10^{-3}$, over the same 200 trials at seed 0 as Table 13, M2 falls from $1,889,539 \pm 64,528$ to $1,398,814 \pm 32,966$ shots and M3 from $1,320,830 \pm 57,944$ to $904,382 \pm 20,201$, savings of 26% and 31%. In all 200 trials the generator returned was still the exact maximiser. Note that $\tau = \text{gap}/4$ uses the exact gap, so this inherits the oracle-knowledge caveat of Sec. 10; a deployment would set τ from a chemical accuracy target instead. Measured on one case only.

13 Parameter sensitivity

13.1 What removing the two free parameters was worth

The previous revision evaluated M1 at a target radius of $\text{gap}/4$ with no task-level justification, and evaluated the BAI methods on a geometric schedule with shrink factor $\gamma = 0.5$. Both have been replaced. The M1 target is now $\text{gap}/2$, the loosest common radius at which a non-adaptive estimate separates the winner from the field, so that all three methods answer the same question; this divides the M1 column by

four. The BAI methods are evaluated in the schedule-free limit; a coarse geometric schedule overshoots each elimination point and overpays.

The schedule change is not cosmetic. Under $\gamma = 0.5$, M3 cost more than M1 on three H_4 rows. In the exact limit it does not, on any row (Sec. 10), so that ordering is a property of the limit rather than of the method; under a coarse schedule, and under finite shots (Sec. 12.1), it can fail.

Table 17: Convergence of a geometric radius schedule onto the exact limit, H_4 side 1.0, HF.

γ	0.5	0.7	0.85	0.97	exact limit
M2: BAI-FC-IG	1,494,274	1,267,755	1,080,113	885,115	854,629
M3: BAI-FC-UG	928,366	872,427	712,219	584,142	566,951

For M2 the limit is the closed form

$$M_{M2} = \sum_i \left(\frac{z_\delta \sum_\alpha \sigma_{i\alpha}}{\Delta_i/2} \right)^2, \quad (28)$$

with Δ_i replaced by the gap for the leader. For M3 the trajectory has one breakpoint per distinct Δ_i and the allocation of Eq. (21) is evaluated at each.

13.2 The confidence level does not change the comparisons

Every planning bound scales with z_δ^2 , so every ratio in Table 6 is invariant under the family-wise failure probability. Verified for $\delta \in \{0.2, 0.05, 0.01, 0.001\}$: the H_4 side 1.0, HF ratios stay at 2.63 and 1.51 to better than one part in 10^3 , while the absolute costs move by about a factor of two.

This invariance is a property of the bounds and does not extend to realised cost. Once a schedule and finite shots are introduced, z_δ changes where each elimination fires and therefore how much each method overshoots, and the two methods are not affected equally: Table 16 gives realised M2/M3 ratios of 1.60, 1.36, 1.77 and 1.50 at $z_\delta = 1.39, 1.80, 2.10$ and 3.10 on one H_4 row — a spread of 1.3, with no monotone pattern. What δ does to the realised error rate is treated in Sec. 12.5.

14 Routine independence

Every number above comes from one first-fit grouping with one insertion order. First-fit is deterministic given an order, but the order is a free convention, so the conclusions have to be tested against it. The insertion order is now a parameter: three deterministic conventions — by decreasing weight (**weight**, the default used everywhere else), by increasing weight (**reverse-weight**), and lexicographic — together with a **random** family under seeds 0, 1, 2, 3. The H_4 sweep uses all seven; the LiH sweep uses the three deterministic orderings and **random** seed 0; the H_2O sweep uses the four orderings with **random** seed 0 and varies the parent grouping only, leaving M2’s and the baseline’s individual groupings at the default. Every combination is recorded per case in `*grouping_routines.csv`.

Table 18: Spread of the method ratios across groupings of the same problem. H_4 uses seven groupings and LiH four, each recomputing the baseline and all three methods. The H_2O row varies the parent grouping only, over four orderings.

Case	Contexts	M1/M2	M2/M3	M1/M3	Baseline/M3
H_4 , side 1.0, HF	49–61	1.91–2.63	1.51–1.97	3.35–4.55	8.47–12.27
LiH, $R=3.0$, HF	980–1,127	2.58–3.12	1.79–2.11	5.22–5.58	10.68–12.68
H_2O str., $R=1.75$, HF	2,356–2,870	—	0.74–1.30	174–262	—

On H_4 and LiH the ratios move by at most a factor of 1.45 and the ordering of the methods is identical under every grouping tested. Those two systems are routine-independent for the quantities in the table.

H_2O is not. The M3 cost varies by a factor of 1.77 across parent groupings, from 1.34×10^7 to 2.37×10^7 , and the M2/M3 verdict changes sign: M3 is cheaper under two of the four orderings and more expensive under the other two. **The M2 versus M3 comparison on H_2O is therefore not resolved.** It lies inside the spread induced by an arbitrary convention, and no verdict should be claimed in either direction — neither the planning-bound result that M2 wins on the stretch HF row, nor the finite-shot result of Sec. 12 that M3 does.

This applies to all four H_2O rows, not only the swept one. The three unswept rows give $\text{M2/M3} = 1.13, 1.03$ and 1.01 , every one of which lies inside the measured 0.74 to 1.30 spread. They are reported in Table 6 because they are what the fixed convention gives, but none of them is far enough from unity to survive the convention being changed, and the sweep that would settle them has not been run. The M1 versus M3 comparison on the same rows is never in doubt, at 174 to 262 on the swept row and 7.8 to 192 across the four.

15 Within-context correlation

Assumption (iii) of Sec. 7 is removed here. The covariance of two generators’ fragments inside one context follows from variances alone,

$$\text{Cov}(F_{i\alpha}, F_{j\alpha}) = \frac{1}{2} [\text{Var}(F_{i\alpha} + F_{j\alpha}) - \text{Var}(F_{i\alpha}) - \text{Var}(F_{j\alpha})], \quad (29)$$

and sampling from those matrices makes the simulation exact for the shared methods as well. The matrices are formed per context, symmetrised, and factorised through an eigendecomposition with negative eigenvalues clipped to zero, which keeps the draw well defined when a context carries almost no variance.

The sampler has been verified. On all three cases below, every context covariance is positive semidefinite to rounding (most negative eigenvalue -5×10^{-16}), every diagonal entry equals the corresponding $\sigma_{i\alpha}^2$ exactly, and the empirical marginal variance of each generator over 2×10^4 draws lies between 0.976 and 1.023 of its analytic value. The anomaly reported below is therefore not a marginal-sampling defect.

15.1 The covariance-aware elimination rule, and a sign defect in it

Where the *marginal* rule of Eq. (9) compares $|\hat{g}_i|$ and $|\hat{g}_j|$ at their individual radii, the *pairwise* rule compares each pair at the radius of the combination the comparison actually depends on. Because the comparison is between absolute values, that combination is $\hat{g}_i - s_{ij}\hat{g}_j$ with $s_{ij} = \text{sign}(\hat{g}_i\hat{g}_j)$, so i is eliminated when some active j satisfies

$$|\hat{g}_j| - |\hat{g}_i| > z_\delta \sqrt{V_{ij}}, \quad V_{ij} = \sum_\alpha \frac{1}{n_\alpha} \text{Var}(F_{i\alpha} - s_{ij}F_{j\alpha}), \quad (30)$$

with the same z_δ of Eq. (20). When the two fragments are correlated in the direction that helps, V_{ij} is below $\text{Var}(\hat{g}_i) + \text{Var}(\hat{g}_j)$ and the rule fires earlier.

The sign factor matters and was missing from the previous revision, which used $\text{Var}(\hat{g}_i - \hat{g}_j)$ unconditionally. That is not only the wrong combination for an opposite-sign pair; it is not even well defined, because the sign of a gradient depends on the arbitrary phase of the Hartree–Fock orbital it excites from. Only $s_{ij} \text{Cov}(\hat{g}_i, \hat{g}_j)$ is phase-invariant, and Eq. (30) uses exactly that. The defect is corrected in the numbers below.

Table 19: M3 under correct joint noise, 250 trials per row at seed 0, with the corrected pairwise rule. Uncertainties are one standard error; brackets are Wilson 95% intervals on the correct-selection rate.

Case	Noise model, rule	Shots	Correct [95% CI]	vs indep.
H ₄ , side 1.0, HF	independent, marginal	1,334,743 ± 51,258	100.0% [98.5, 100]	—
	correlated, marginal	1,348,697 ± 49,914	100.0% [98.5, 100]	1.01
	correlated, pairwise	757,292 ± 65,390	98.8% [96.5, 99.6]	0.57
H ₄ , side 1.0, CISD	independent, marginal	148,219 ± 1,989	100.0% [98.5, 100]	—
	correlated, marginal	146,140 ± 1,808	100.0% [98.5, 100]	0.99
	correlated, pairwise	88,995 ± 2,630	100.0% [98.5, 100]	0.60
H ₄ , side 2.0, HF	independent, marginal	544,228 ± 10,397	100.0% [98.5, 100]	—
	correlated, marginal	6,475,180 ± 793,419	75.6% [69.9, 80.5]	11.90
	correlated, pairwise	9,070,121 ± 925,919	51.2% [45.0, 57.3]	16.67

15.2 Two benign cases and one that is not

On both H₄ side 1.0 rows, modelling the correlation changes the cost by 1 to 2%, and the covariance-aware rule then cuts M3’s realised cost by 40 to 43% with no meaningful loss of accuracy. This is the mechanism Part II proposes for the decisive stage, appearing here as a direct reduction in shots rather than as the outcome of an optimisation over how to split coefficients.

H₄ side 2.0 HF behaves differently and is **still not explained**. Correct selection falls to 76% under the marginal rule and 51% under the pairwise one, far outside sampling noise, with costs rising more than tenfold. The sign correction above does not recover it, and neither does correcting the confidence factor over pairs rather than over generators. Three properties distinguish this case from the two benign ones, and any explanation has to account for them:

- **The runner-up is exactly degenerate.** D 0,3 → 4,7 and D 1,2 → 5,6 both have $|g| = 0.298387$, so there is no unique second arm and $|\mathcal{A}| = 3$.
- **The leading pair is anticorrelated**, with $\text{Cov} = -1.4 \times 10^{-2}$ against variances of 6.7×10^{-2} and 9.6×10^{-2} . The previous revision reported the diagnostic ratio $\text{Var}(\hat{g}_1 - \hat{g}_2)/[\text{Var}(\hat{g}_1) + \text{Var}(\hat{g}_2)]$ as 0.82 and concluded that the difference is easier to resolve than independence implies. *That was the wrong quantity*: the ratio so defined is 1.168 here, and 0.832 is the value for the sum, which is the relevant combination only because the pair has opposite signs. The corresponding ratios on the two benign rows are 0.929 and 1.084. The earlier claim that the obvious explanation is “contradicted by direct measurement” is withdrawn.
- **The gap is narrow relative to the field**, $\text{gap}/|g_{(1)}| = 5\%$ with $\max_i \Delta_i/\text{gap} = 18.9$, so a 17% error in the assumed variance of the decisive comparison has room to matter. H₄ side 1.0 CISD is also anticorrelated, at 1.084, but its gap is 83% of $|g_{(1)}|$ and nothing goes wrong.

A further complication is specific to this system and is recorded in Sec. 19: the Hartree–Fock orbital phases for square H₄ are not reproducible across SCF starting guesses, so the *relative signs* of the gradients — and hence every covariance between two of them — differ between runs, while all magnitudes agree. Nothing in the planning bounds or the marginal rule depends on those signs. The correlated model is the one place they enter.

Status. The independent noise model of Sec. 12 remains the one used for every reported number. The correlated results in Table 19 should be treated as *unverified* on H₄ side 2.0 until the anomaly is diagnosed.

16 Conclusions from the current data

Table 20 gives the evidence level behind each item. Items marked there as single-system or bound-only should not be generalised without the work in Sec. 18.

1. The four constructions are implemented as separate modules that save their FC groups, fragment variances and elimination histories, and are evaluated at one criterion with no free schedule or precision parameter. The validation gate passes on all nine cases to 4×10^{-15} Ha.
2. **On the planning bounds M3 is cheaper than M1 in every case**, by 1.1 to 192 (Table 6). *Observed, not proved* — the sum-of-maxima accounting of Eq. (14) makes the obvious argument incomplete — and *bound-level only*: on realised cost M3 loses to M1 on both H₄ CISD rows (Sec. 12.1). All M1 columns assume M1 is handed the exact gap.
3. **M3 is cheaper than M2 on H₄ and LiH; on H₂O the comparison is unresolved.** M3 wins eight bound-level rows of nine, but all four H₂O rows fall between 0.86 and 1.13, inside the 0.74–1.30 range that changing the grouping routine produces. No verdict should be claimed on any H₂O row in either direction, including the finite-shot one. The sweep itself covers H₄ side 1.0 HF and LiH only.
4. **M2 is not always cheaper than M1**, winning seven bound-level rows of nine and three of seven on realised cost. The counterexamples are the wide-gap rows where static estimation is already cheap; LiH has a comparably wide gap and does not invert, so the gap alone is not the explanation. This comparison is also the most sensitive to M1’s oracle gap (App. D).
5. **The two mechanisms are substitutes.** Their interaction against the no-sharing baseline is below unity in all nine cases, between 0.57 and 0.96: elimination removes the situation that sharing exploits.
6. **Elimination is the dominant mechanism in seven rows of nine.** Elimination alone is worth 1.27 to 252.97 against the baseline, sharing alone 1.11 to 3.35; on both H₄ CISD rows sharing wins (2.19 against 1.27, 3.11 against 2.08). The sharing factor is smallest on all four H₂O rows, which with four systems is a correlation with system rather than a trend with size, and refers to planning bounds only.
7. **Under finite shots M3’s cost inflates less than M2’s on the three largest cases.** M2 inflates by 2.2 to 3.2 everywhere; M3 by 2.3 to 3.0 on the four H₄ rows but only 1.40 to 1.66 on LiH and both H₂O rows, where the M2/M3 ratio roughly doubles (200 trials per row, separations far outside the standard errors). *No mechanism is established*: the breadth-dominance explanation fails on the two H₄ CISD rows, and an unseparated part of every inflation figure is discretisation of the $\gamma = 0.5$ schedule rather than noise.
8. **The confidence convention does not control the selection-error rate.** Sweeping δ from 0.05 to 0.99 moves the realised rate only within 0–0.7%, and 300 trials bound it only below 1.3%. Inverting Eq. (26) for 5% gives 10 to 12% in simulation, because that expression ignores generators eliminated wrongly earlier in the run. An optimum sits near $z_\delta \approx 1.8$ –2.1 on the one case tested. Invariance of the method ratios to z_δ holds for the *planning bounds only*: the realised M2/M3 ratio moves over 1.36 to 1.77.
9. **Conclusions 2 to 6 are routine-independent at bound level on H₄ side 1.0 HF and LiH**, the two rows swept: ratios move by at most 1.45 and no ordering changes. The sweep covers no other H₄ row, no H₂O row beyond one parent grouping, and neither the finite-shot nor the correlated-noise study.
10. **Within-context correlation is a saving on two cases and unexplained on a third.** On both H₄ side 1.0 rows it changes cost by 1 to 2% and the covariance-aware rule then cuts M3’s realised cost by 40 to 43%. On H₄ side 2.0 HF accuracy falls to 75.6% and 51.2%. The sampler is verified correct marginally and a sign defect in the pairwise rule has been fixed, so neither accounts for it;

the case has a doubly degenerate runner-up, an anticorrelated leading pair and a narrow gap. *Those results are unverified* and the independent noise model remains the reported one.

11. The physics of the previous revision is confirmed; its cost model is not reproducible and its M1/M2 ratios, from 0.28 to 627, should not be used (App. A).
12. All numbers are single-step, single-state diagnostics counting shots only. Nothing here addresses a full ADAPT trajectory, nor circuit-level cost, where M2 and the baseline need 3.5 to 6.1 times as many distinct measurement settings (Table 3).

Table 20: Evidence behind each conclusion. “Bound” is the oracle planning model, “finite-shot” the surrogate of Sec. 12, “swept” the grouping-routine sweep of Sec. 14.

#	Claim	Bound	Finite-shot	Swept
2	M3 < M1	9 of 9	5 of 7	H ₄ 1.0 HF, LiH
3	M3 < M2 (H ₄ , LiH only)	8 of 9	7 of 7	H ₄ 1.0 HF, LiH
4	M2 < M1	7 of 9	3 of 7	H ₄ 1.0 HF, LiH
5	Substitutes, interaction < 1	9 of 9	—	H ₄ 1.0 HF, LiH
6	Elimination dominates	7 of 9	—	H ₄ 1.0 HF, LiH
7	M3 inflates less on large cases	—	7 cases, 200 trials	no
8	δ does not control error	—	1 case, 300–500 trials	no
10	Covariance-aware rule	—	3 cases, 250 trials, 1 unverified	no
12	Distinct-setting counts	4 systems	—	no

17 Interface with Parts II and III

Part II improves M3 by using measured variances and covariances to optimise overlapping universal operators and coefficient splitting. Sec. 10.2 gives that work a sharper target, but Sec. 10.4 also raises a caution that should be settled before it is built.

The mechanism Part II proposes for the decisive stage is a pairwise-variance identity [AUTHOR: cite Part II]: best-arm identification only requires the pairwise differences to be resolved, and

$$\text{Var}(\hat{g}_i - \hat{g}_j) = \sum_{\alpha} \frac{1}{n_{\alpha}} \text{Var}(F_{i\alpha} - F_{j\alpha}), \quad (31)$$

so the covariance that helps enters through co-measurement inside a shared context, not through the two generators sharing Pauli products. That premise was tested on the finalist pair of each system, inside the parent contexts, by comparing the cost of resolving the difference alone against the cost of resolving both generators to the same radius.

Table 21: Separating the two finalists inside the shared parent contexts: resolving only their difference against resolving each of them, both to radius gap/2. All sums are over parent contexts, so these are the S^{par} of Sec. 10.2; S_{i-j}^{par} refers to the single observable $C_i - C_j$.

Case	S_i^{par}	S_j^{par}	S_{i-j}^{par}	Resolve both	Resolve difference
H ₄ , side 1.0, HF	1.921	1.731	2.455	554,857	499,910
LiH, $R=3.0$, HF	2.170	2.180	3.623	48,712	67,577
H ₂ O eq., $R=0.9572$, HF	7.263	12.173	17.803	1,002,360	1,581,034
H ₂ O str., $R=1.75$, HF	6.239	7.794	13.005	10,636,302	18,049,318

Resolving the difference is cheaper only on H₄, by a factor of 1.11; on LiH and both H₂O rows it is more expensive, by factors up to 1.7. The arithmetic is simple. Writing S for S^{par} throughout, resolving two generators separately costs $S_i^2 + S_j^2$ in units of ϵ^{-2} , while resolving one difference observable costs

S_{i-j}^2 , so the difference wins only when $S_{i-j} < \sqrt{S_i^2 + S_j^2}$, which for comparable arms demands roughly 30% cancellation. The measured cancellation, $1 - S_{i-j}/(S_i + S_j)$, is 33% on H_4 but only 17% on LiH and 7% on H_2O stretch. It is tempting to read this against the finalist overlaps of Table 10, but the two do not line up: H_4 has zero finalist overlap and the *largest* cancellation, LiH has 35% overlap and half the cancellation. Sharing Pauli products is evidently not what produces cancellation in the difference — which is consistent with the premise stated above, that the useful covariance comes from co-measurement inside a context rather than from shared support, but leaves the size of the cancellation unexplained by anything measured here.

This does not refute Part II, whose other mechanisms — coefficient splitting over an overlapping cover, empirically estimated context covariances, and adaptive choice of the next context to measure — are untouched by this test. It does mean the pairwise-difference mechanism should not be assumed to pay, and that whether cancellation is available must be measured on the intended benchmark before the variance-aware machinery is designed around it. Sec. 10.2 adds a second caution: that mechanism acts on the depth term, which is only 12% of the M3 cost on LiH and about 39% on H_2O stretch, so its total effect is bounded even where it does pay.

Part III changes the ADAPT trajectory so that the commutators first become symmetry-filtered and tapered. When Part III is active, the Part I object C_i is replaced by

$$C_{i,\text{tap}}^{\parallel,k} = \text{Taper}_{\mathcal{R}_k}(\Pi_{\mathcal{R}_k}([H, G_i])), \quad (32)$$

and then M1, M2, or M3 is applied to these reduced operators. Part I tables do not include this reduction.

18 Remaining work in Part I

1. **Diagnose the H_4 side 2.0 correlated-noise failure** (Sec. 15). The sampler’s marginals and positive semidefiniteness are verified and a sign defect in the pairwise rule is fixed, so neither explains it. The next candidates are the exactly degenerate runner-up, which leaves no unique second arm, and the interaction of the anticorrelated leading pair with the elimination threshold. Until this is settled the correlated model cannot be used for reported numbers.
2. **Extend the routine sweep to the remaining H_2O rows.** One row has been swept and it dissolved a conclusion. The other three have not, nor have the individual groupings that M2 and the baseline depend on, nor the finite-shot and correlated-noise studies, which were run under one routine throughout. This is the single largest gap in the evidence base.
3. **Test the confidence convention beyond one case.** The optimum at $z_\delta \approx 1.8$ to 2.1 rests on H_4 side 1.0 HF alone. It should be measured on LiH and at least one H_2O row before any convention is adopted, because choosing it changes every absolute number and all tables would have to be regenerated.
4. **Test good-enough selection beyond one case** before making it the default stopping rule. The 30% saving is measured on H_4 side 1.0 HF only, and the tolerance used, $\text{gap}/4$, is derived from the exact gap; a deployable version needs a tolerance defined from a chemical accuracy target instead.
5. **Simulate the two H_2O CISD rows**, the only cases missing from Table 13. They are the rows where the planning bound is largest, so they are the ones where a realised-versus-bound discrepancy would matter most.
6. **Separate schedule discretisation from measurement noise** in the finite-shot inflation. The $\gamma = 0.5$ schedule overpays by 1.6 to 3.8 at the planning level with no noise at all, so the reported inflations of 1.4 to 3.2 are a mixture of the two that this study cannot decompose, the noiseless and

noisy runs having different stopping rules. Running the surrogate across several γ would settle it and would also show how much of the reported cost a tuned schedule recovers.

7. **Measure the excluded cost components.** The shot metric prices neither circuit depth nor the number of distinct measurement settings, and the latter differs by 3.5 to 6.1 times between the method families (Table 3). Until that is priced, the M2-versus-M3 comparison is incomplete even where the shot counts are resolved.
8. **Complete the test suite.** Sec. 19 lists what is covered and what is not; the highest-value gaps are the brute-force commutator and variance checks, an adversarial search for a counterexample to $M_{M3} \leq M_{M1}$, and a shot-level sampling check of the Gaussian surrogate.

19 What is tested, and what is not

Covered. The suite of 26 unit tests pins the conventions that were previously wrong and the pieces with closed forms: the computational-basis ordering and big-endian convention; the qubit Hamiltonian against the PySCF RHF energy on the Hartree–Fock determinant and against the PySCF FCI energy in the target sector, on every case, as a hard gate; the top gradient against a finite-difference energy derivative; the context-shot allocation against Eq. (18) for a single gradient; and the fragment decomposition summing back to the gradient operator. Alongside these, `verify_part1.py` runs 68 checks on the published results: that each reported total is the sum of the file that claims to explain it, that the validation gate passed, that the allocation meets its confidence target, that $M_{M3} \leq M_{M1}$, that all methods agree on the selected generator, that repeated runs agree exactly, that the ratios are invariant to δ , and that a geometric radius schedule converges onto the exact limit. All pass.

Not covered. The following would materially raise confidence and have not been done. They are listed in rough order of value.

1. **Brute-force commutators and variances.** Every $C_i = [H, G_i]$ should be checked against a dense matrix commutator on H_4 , every gradient (not only the top one) against finite differences, and every fragment variance against a dense $\langle F^2 \rangle - \langle F \rangle^2$. The pipeline is currently validated at its ends but not term by term.
2. **Grouping validity.** That every pair within a context commutes, that the contexts partition $\mathcal{B}_0$ exactly once, and that each restricted context $\mathcal{M}_\alpha \cap \mathcal{B}_t$ is still fully commuting, are asserted by construction rather than tested.
3. **Shot-level sampling of the joint distribution.** The per-context covariances and the marginals they generate are now checked (Sec. 15), but the joint draw has not been compared against sampling actual measurement outcomes, which is what would settle the H_4 side 2.0 anomaly.
4. $M_{M3} \leq M_{M1}$ **adversarially.** The ordering is checked on nine cases and argued informally; random multi-constraint instances should be searched for a counterexample to the sum-of-maxima aggregation, since finding one would turn conclusion 2 from an observation into a bounded claim.
5. **The Gaussian surrogate against shot-level sampling.** On H_4 , contexts could be diagonalised and bitstrings sampled, and the empirical mean and variance of $\hat{g}_i$ compared with the normal model. Assumption (i) of Sec. 7 is currently untested.
6. **Allocation optimality.** The fixed-point solver agrees with a general convex solver on the small instances tried; the KKT residual should be reported as a matter of course, and multi-constraint instances tested systematically.

7. **Analytically known answers.** H_2 in a minimal basis has gradients, commutators and fragment variances obtainable by hand; a two-arm synthetic BAI problem with a known gap would pin the elimination order and Eq. (28).
8. **Limiting cases and symmetry.** A single-generator pool must give $M_1 = M_2 = M_3$; a vanishing gap must diverge as $1/\text{gap}^2$; gradients that vanish by symmetry must fall below threshold; results must be invariant under relabelling qubits or permuting the pool; the pool sizes 26, 92, 140 should be checked against the combinatorial formula and every generator against anti-Hermiticity.
9. **Regression pinning.** Every table entry should be pinned to a stored value with a commit hash, so that any change is caught rather than discovered.

One reproducibility limit is known and not a test gap. The Hartree–Fock orbitals of square H_4 are determined only up to the sign of each molecular orbital, and different SCF starting guesses return different signs. Gradient *magnitudes*, fragment variances and every quantity in the planning bounds are unaffected, and repeated runs from the same starting guess agree exactly. The relative *signs* of two gradients, and hence the covariance between them, are not reproducible unless the phases are fixed. Only the correlated noise model of Sec. 15 depends on them. Fixing a phase convention — for instance, requiring the largest-magnitude coefficient of each orbital to be positive — would remove this, and should be done before that model is used.

The implementation should be regarded as validated at the level these tests reach and no further. Two defects of the previous revision were silent and were caught only by end-to-end energy comparison (App. C); the list above is written in that light.

20 Reproduction

```
python -m pip install -r requirements.txt
python scripts/run_part1_methods.py --all --out runs
python scripts/run_finite_shot.py --case H4_square_eq_side1p0_HF \
    --trials 200 --out runs
python -m pytest tests -q
python scripts/verify_part1.py --runs runs
```

Each case writes a per-method summary together with the FC groups, fragment standard deviations and BAI histories behind it, and records the interpreter and package versions with every run. `verify_part1.py` checks the published results rather than the implementation: for every case it confirms that the reported total is the sum of the file that claims to explain it, that the validation gate passed, that the allocation meets its confidence target, that $M_{M3} \leq M_{M1}$, and that all methods agree on the selected generator; against the live code it confirms that the fragment decomposition sums back to the gradient, that repeated runs agree exactly, that the ratios are invariant to the confidence level, and that a geometric radius schedule converges onto the exact limit used in the tables. All 68 checks pass, alongside 26 unit tests. Sec. 19 says what those tests do and do not cover.

Case identifiers accepted by `--case`:

LiH_R3p0_HF	
H4_square_eq_side1p0_HF	H4_square_eq_side1p0_CISD
H4_square_stretch_side2p0_HF	H4_square_stretch_side2p0_CISD
H20_eq_HF	H20_eq_CISD
H20_stretch_HF	H20_stretch_CISD

Determinism and seeds. The planning bounds contain no randomness: given a case and a grouping routine, every number in Secs. 10–11 is reproduced exactly on re-running, and `verify_part1.py` asserts this. The finite-shot tables use `numpy` default-RNG streams seeded at 0 for Table 13 and per-entry seeds recorded in the run summaries for Tables 15–16 and 19. The grouping-routine sweep uses the `random` orderings at seeds 0 to 3.

Numerical settings. PySCF defaults are used for the SCF convergence threshold and initial guess, except that square H_4 needs the phase convention noted in Sec. 19 to make covariances reproducible. The CISD state is the lowest eigenpair of the Hamiltonian restricted to the ≤ 2 -excitation space, taken with a dense symmetric eigensolver below 2^{12} basis states and with `scipy.sparse.linalg.eigsh` at tolerance 10^{-12} above it. The convex-solver cross-check of Sec. 5 used SLSQP on randomly generated instances with up to eight arms and twenty contexts, agreeing to better than one part in 10^6 .

Environment and provenance. [AUTHOR: record the repository URL and the commit hash that produced these tables, the contents of `requirements.txt`, the PySCF, OpenFermion, NumPy and SciPy versions, and the wall-clock time and hardware for the H_2O CISD rows.] Every run already writes these versions into its summary file; they are simply not yet transcribed here, and the repository referred to above does not yet have a public location.

A Revision history and relationship to the previous revision

Note on the finite-shot tables. The finite-shot study was regenerated for this revision: every row now uses 200 trials at seed 0, where the H_2O rows previously used 40, and standard errors and Wilson intervals are reported throughout. Totals therefore differ from the previous revision at the few-per-cent level of the run-to-run scatter (for instance LiH M3, 882,018 before and $893,509 \pm 19,804$ now). The correlated-noise table was also regenerated after the sign correction described in Sec. 15. The breadth metric of Table 8 changed definition: it was previously the share paid by the top two arms for M2, which gave 53.8% on H_4 side 1.0 HF against 3.0% under Eq. (23).

What this revision changed. All numbers are produced by a single from-geometry implementation in which each method is a separate module that saves the fully commuting groups and the fragment variances behind every table entry. Every run passes the validation gate of Sec. 6. All nine cases are complete, H_2O included. Both free parameters of the previous revision have been removed: M1 is evaluated at the target radius that matches the winner-identification task, and the BAI methods in the schedule-free exact limit rather than on a geometric radius schedule. Two further pieces were added: the no-sharing baseline of Sec. 11, which completes the two-by-two the methods span, and the finite-shot surrogate of Sec. 12, which retracts one conclusion of the earlier revision and shows the confidence convention to be far stricter than the stated failure probability requires.

The physics of the previous revision is confirmed. Universal Pauli supports, gradient values, gaps and selected generators are reproduced to six or more digits. The present code was written from the molecular geometry upward rather than derived from the previous revision’s scripts, and shares no module with them; it is *not* an independent reimplementaion in the stronger sense, since it was written by the same project against the same conventions, so the agreement is evidence that the physics layer is correct and not that two independent groups agree. What changed is the cost model, for three reasons.

First, the fully commuting grouping. All rows now use one deterministic greedy routine; the previous revision’s H_4 and H_2O rows used a different partitioning, giving 356 and 9,874 contexts where greedy gives 53 and 2,366. Cost grows with the number of fragments an operator is split across, so this alone is a factor of several.

Second, the previous M1 column was not a single model. For LiH it is reproduced within 8% by a uniform per-context allocation at one standard error sized to a *typical* rather than worst-case generator (1,929,181 against the reported 2,088,380), which guarantees the target precision for about half the gradient vector rather than all of it. For H_4 no combination of grouping, allocation and confidence

convention reproduces the reported 182,282,619: uniform allocation at worst-case variance with the previous revision’s own 356 contexts lands near 3×10^7 . That entry also exceeds its own no-sharing baseline of 11,326,894 by a factor of 16. That is not strictly impossible — a shared grouping fragments each gradient, so with a fragmentation penalty $f_i > 1$ an allocation over shared contexts can in principle cost more than measuring each generator in its own grouping, and a bound of the form $\max_i f_i$ times the baseline is the most that can be claimed in general. But the f_i measured here are between 1.07 and 3.62 (Table 9), and the present M1 is *below* its baseline on all nine rows, so a factor of 16 above it remains unexplained by any mechanism in the model.

Third, the allocation itself. On LiH, solving Eq. (21) costs about 17 times fewer shots than spreading the same total uniformly over the 980 parent contexts, at the same one-standard-error target and the same $z_\delta = 1$. At one standard error the present M1 is *cheaper* than the previously reported value, 1,191,917 against 2,088,380, despite covering the worst-case generator rather than a typical one. The larger absolute values in Table 6 come principally from applying a Bonferroni family-wise z over the pool, a conservative convention applied identically to all three methods and therefore invisible in the ratios.

The consequence is that the previous M1/M2 ratios, which ranged from 0.28 to 627, should not be used. They combine an optimistic LiH entry with an unexplained H₄ entry.

B Naming convention for code, files and plots

The method names **M1: FC-UG**, **M2: BAI-FC-IG** and **M3: BAI-FC-UG** must be used consistently in text, code, tables, plots and file names. Output files carry the method codes M1.FCUG, M2.BAIFCIG, M3.BAIFCUG and NOSHARE.

C Implementation status and past defects

The following items of the previous revision’s cleanup list are complete.

1. Separate output files named with the method codes M1.FCUG, M2.BAIFCIG and M3.BAIFCUG.
2. FC groups, fragment standard deviations and BAI elimination histories saved for every number in every table.
3. Deterministic grouping conventions recorded with every run, together with the interpreter and package versions.
4. M3 completed for H₂O, so the headline comparison no longer has missing entries.

Two implementation defects of the previous revision are worth recording, because both were silent. The two-electron integrals were inserted into the spin-orbital tensor in chemist rather than physicist ordering, with the spin pattern (a, b, a, b) instead of (a, b, b, a) and without the one-half prefactor; and the computational basis was indexed little-endian where the sparse-operator convention is big-endian. For H₄ the resulting Hartree–Fock energy was -5.21 Ha against an RHF value of -1.76 Ha. The validation gate described in Sec. 6 exists to make a repeat of this impossible.

D The M1 target radius, if it is varied anyway

M1 has no internal criterion for its own precision, so it is worth recording how the comparison would move if the target were chosen differently. M2 and M3 are unaffected, their targets being fixed by the deficits.

Table 22: M1: FC-UG cost and the M1/M2 ratio against a target radius of gap/d . $d = 2$ is the identification criterion used in Table 6; $d = 4$ was the previous revision’s choice.

d	1	2	4	8	16
H ₄ , side 1.0, HF: M1	561,216	2,244,862	8,979,447	35,917,787	143,671,148
M1/M2	0.66	2.63	10.51	42.03	168.10
H ₄ , side 1.0, CISD: M1	14,911	59,642	238,565	954,260	3,817,039
M1/M2	0.15	0.58	2.32	9.28	37.13

References

- [1] H. R. Grimsley, S. E. Economou, E. Barnes, and N. J. Mayhall, “An adaptive variational algorithm for exact molecular simulations on a quantum computer,” *Nature Communications* **10**, 3007 (2019). <https://doi.org/10.1038/s41467-019-10988-2>
- [2] V. O. Shkolnikov, N. J. Mayhall, S. E. Economou, and E. Barnes, “Avoiding symmetry roadblocks and minimizing the measurement overhead of adaptive variational quantum eigensolvers,” *Quantum* **7**, 1040 (2023). <https://doi.org/10.22331/q-2023-06-12-1040>
- [3] P. G. Anastasiou, N. J. Mayhall, E. Barnes, and S. E. Economou, “How to really measure operator gradients in ADAPT-VQE,” *arXiv:2306.03227* (2023). <https://arxiv.org/abs/2306.03227>
- [4] R. Huang and A. F. Izmaylov, “Quantum Gambling: Best-Arm Strategies for Generator Selection in Adaptive Variational Algorithms,” *arXiv:2509.14917* (2025); *J. Chem. Theory Comput.* (2026). <https://arxiv.org/abs/2509.14917>
- [5] E. Rossi, E. R. Kjellgren, A. F. Izmaylov, S. P. A. Sauer, K. M. Ziemis, and S. Coriani, “Resource-efficient energy-based operator selection in fermionic ADAPT-VQE via exact Hamiltonian transformation,” *arXiv:2606.04786* (2026). <https://arxiv.org/abs/2606.04786>